

Contents

About	1
0.1 Usage	1
0.2 Render book	1
0.3 Preview book	1
1 Mathematical modelling	2
1.1 Systematic Estimation	3

About

This is a *sample* book written in **Markdown**. You can use anything that Pandoc's Markdown supports; for example, a math equation $a^2 + b^2 = c^2$.

0.1 Usage

Each **bookdown** chapter is an .Rmd file, and each .Rmd file can contain one (and only one) chapter. A chapter *must* start with a first-level heading: # A good chapter, and can contain one (and only one) first-level heading.

Use second-level and higher headings within chapters like: ## A short section or ### An even shorter section.

The index.Rmd file is required, and is also your first book chapter. It will be the homepage when you render the book.

0.2 Render book

You can render the HTML version of this example book without changing anything:

1. Find the **Build** pane in the RStudio IDE, and
2. Click on **Build Book**, then select your output format, or select “All formats” if you’d like to use multiple formats from the same book source files.

Or build the book from the R console:

```
bookdown::render_book()
```

To render this example to PDF as a bookdown::pdf_book, you’ll need to install XeLaTeX. You are recommended to install TinyTeX (which includes XeLaTeX): <https://yihui.org/tinytex/>.

0.3 Preview book

As you work, you may start a local server to live preview this HTML book. This preview will update as you edit the book when you save individual .Rmd files. You can start the server in a work session by using the RStudio add-in “Preview book”, or from the R console:

```
bookdown::serve_book()
```

1

Mathematical modelling

Maths is used to make sense of the world around us, and help us to make decisions, in our personal lives and in the worlds of business, politics, science, and so on. This could involve, for example:

- Creating and solving equations.
- Plotting graphs.
- Making calculations.
- Using computer software.

Each of these things could be thought of as creating a **mathematical model**. This chapter will introduce and explore a range of different kinds of model that may be useful, and when they may be encountered in the real world.

Quote

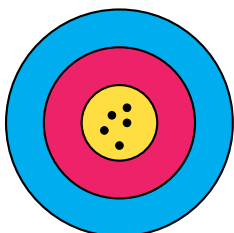
"All models are wrong, but some are useful."

George Box, Statistician

The above quote highlights that mathematical models don't always need to be able to produce 'the *right* answer', and in many situations such a thing is not even possible. Instead, the goal is often a simply a *useful* answer. Since we are entering a world of often *not knowing the true answer*, it is helpful to be learn some related vocabulary:

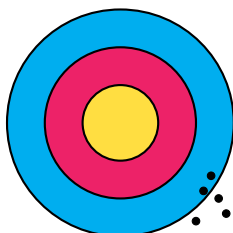
- **Accuracy** relates to how much a measurement or process leads towards the *true value*.
- **Precision** relates to how close repeated measurements or processes are to *each other*.

The results of four different archers aiming for the centre of a target should help illustrate the difference:



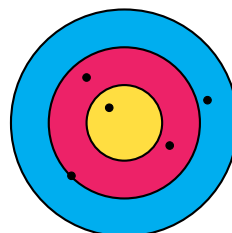
Accurate ✓

Precise ✓



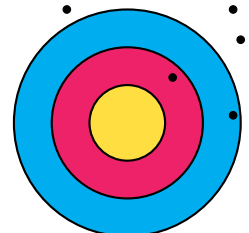
Accurate ✗

Precise ✓



Accurate ✓

Precise ✗



Accurate ✗

Precise ✗

1.1 Systematic Estimation

The 20th century physicist Enrico Fermi was known for performing ‘back of an envelope’ calculations which often gave surprisingly reliable results. Calculations of this type are therefore sometimes called ‘*Fermi estimates*’. The goal of these *systematic estimates* is to quickly get a sense of the *scale* of numerical answers to complex or even impossible questions such as:

Questions

- *How many times might a teacher take a class register in their career?*
- *How much on average does a large supermarket spend on staff wages each week?*
- *How many miles are driven by cars on Scottish roads in a month?*

These calculations might involve using a mixture of *approximated*, *estimated* and *exact* numbers. For example:

- There are **exactly** 60 seconds in one minute.
- There are **approximately** 30 days on average in a month.
- We might **estimate** the mass of a typical family car to be 1000 kilograms.

It would be unlikely that we could come up with *precise* answers using these calculations, but we can aim for *accuracy*. To communicate the main source of any possible lack of accuracy in any such calculation, it is important to state clearly that the answer is based on **assumptions** about the accuracy of any *estimates* made.

Example 1.1A

Problem: A question...

Solution: A solution...

Example 1.1B

Problem: A question...

Solution: A solution...

Now try: **Zeta Higher Apps, Exercise 1.1**